

Memory Efficient Doubly Linked List

–Delete At Position

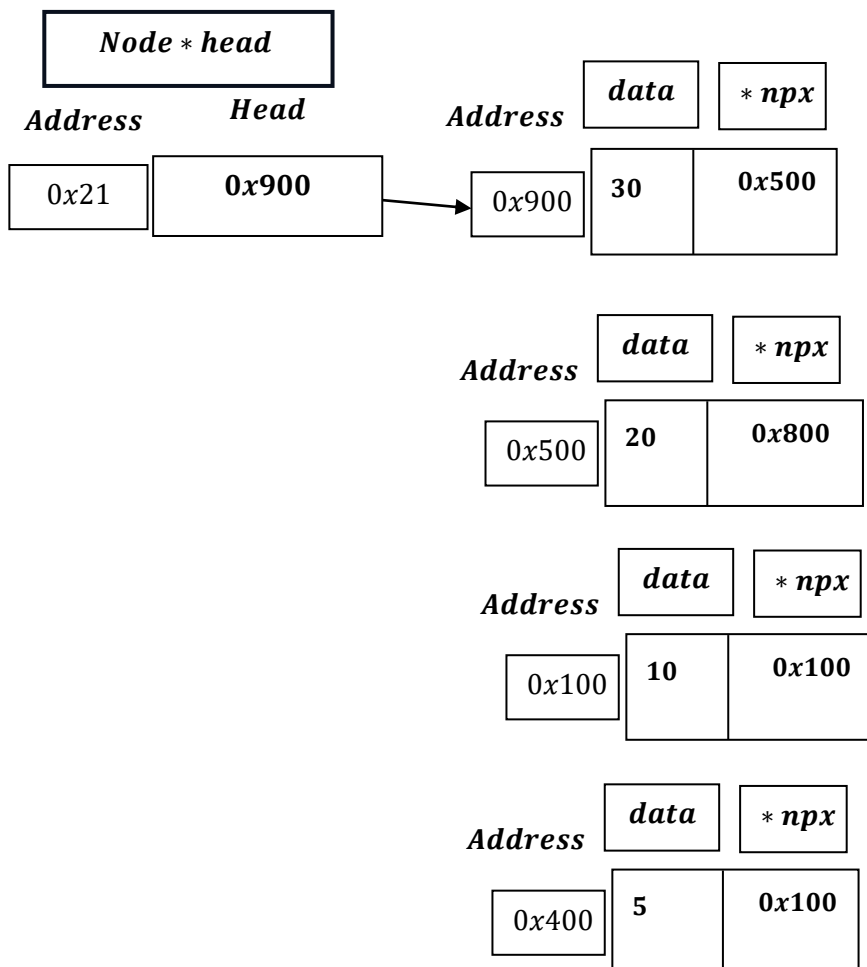
–Program Analysis – Part 2

When next = NULL

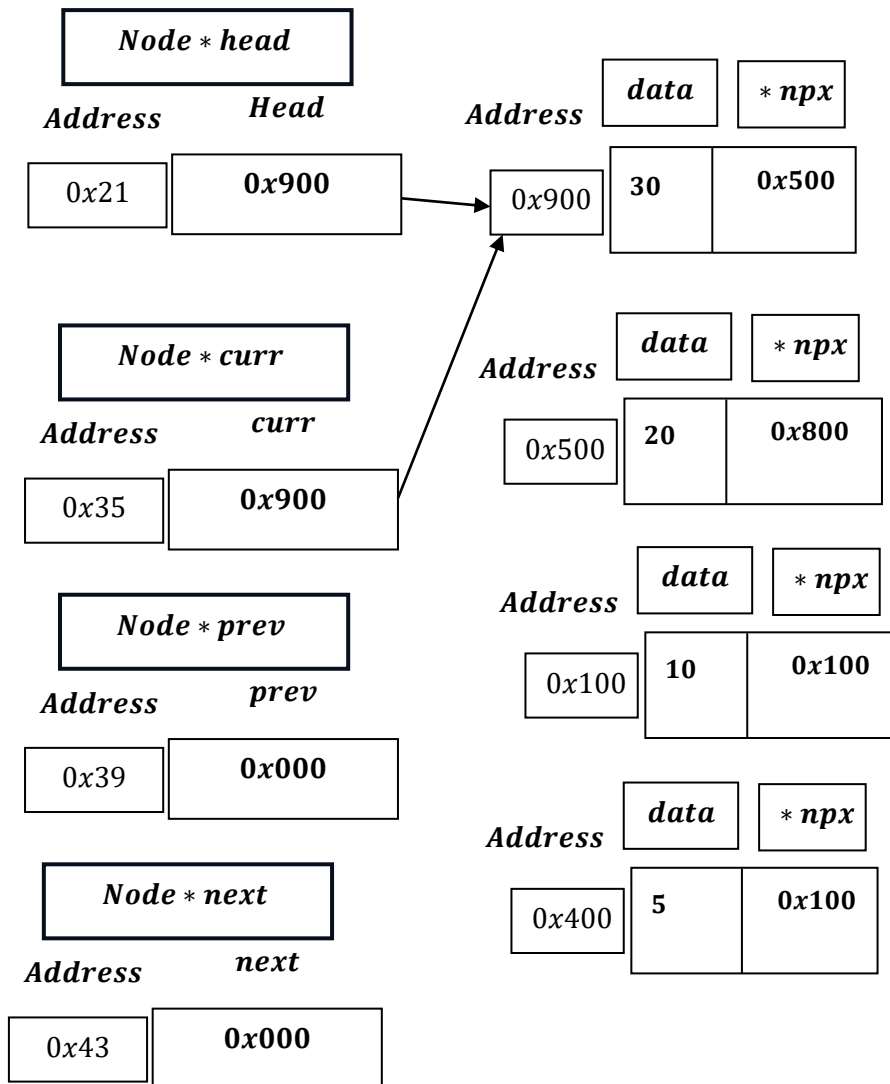
–It occurs when node is deleted at end of the list.

When next $\neq$ NULL

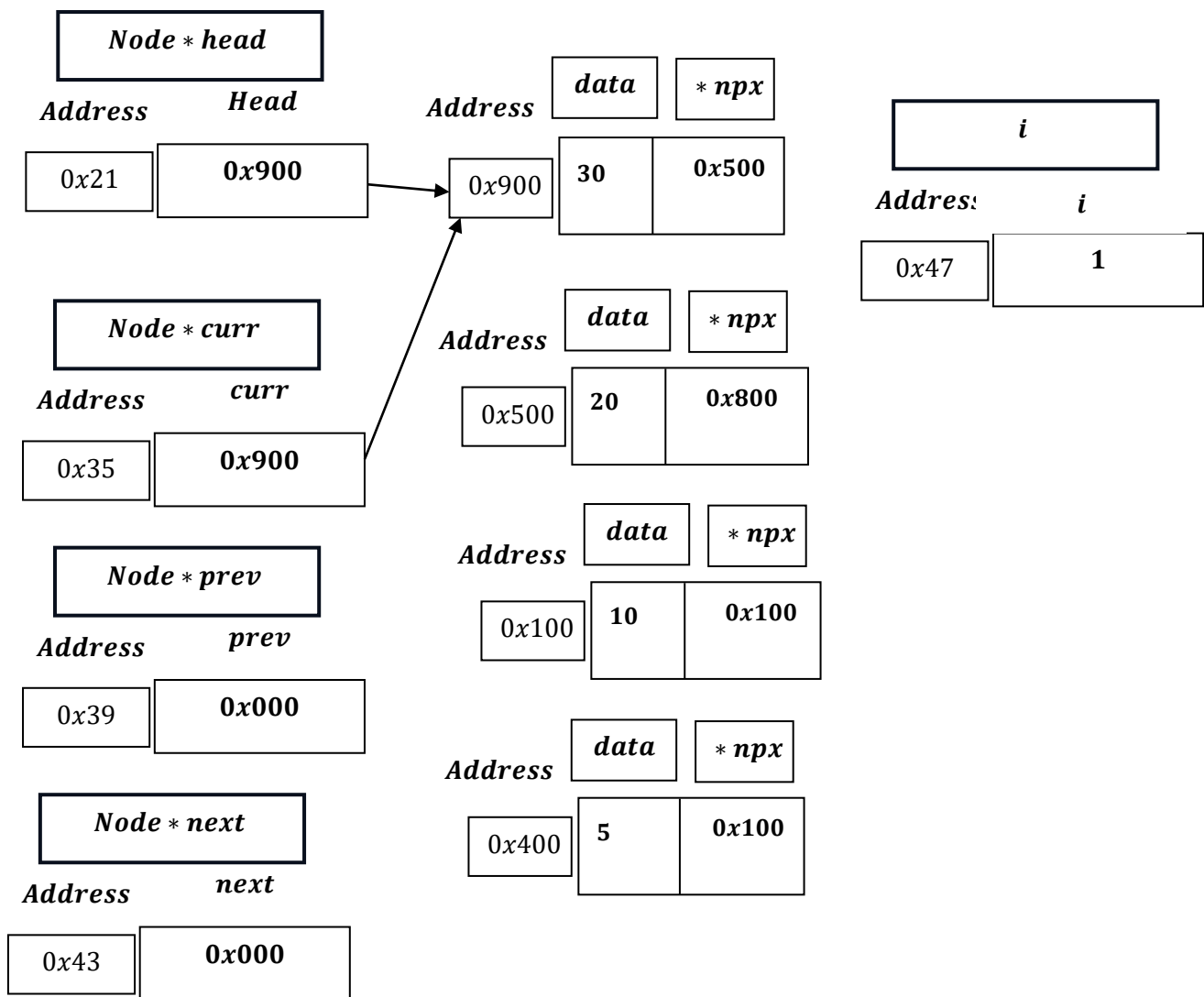
–It occurs when node is not deleted at end of the list.



```
Node *curr = head;
Node *prev = nullptr;
Node *next;
```



```
int i = 1;
```



```
while (curr != nullptr && i < pos)
{
    next = XOR(prev, curr->npx);
    prev = curr;
    curr = next;
    i++;
}
```

pos: 4.

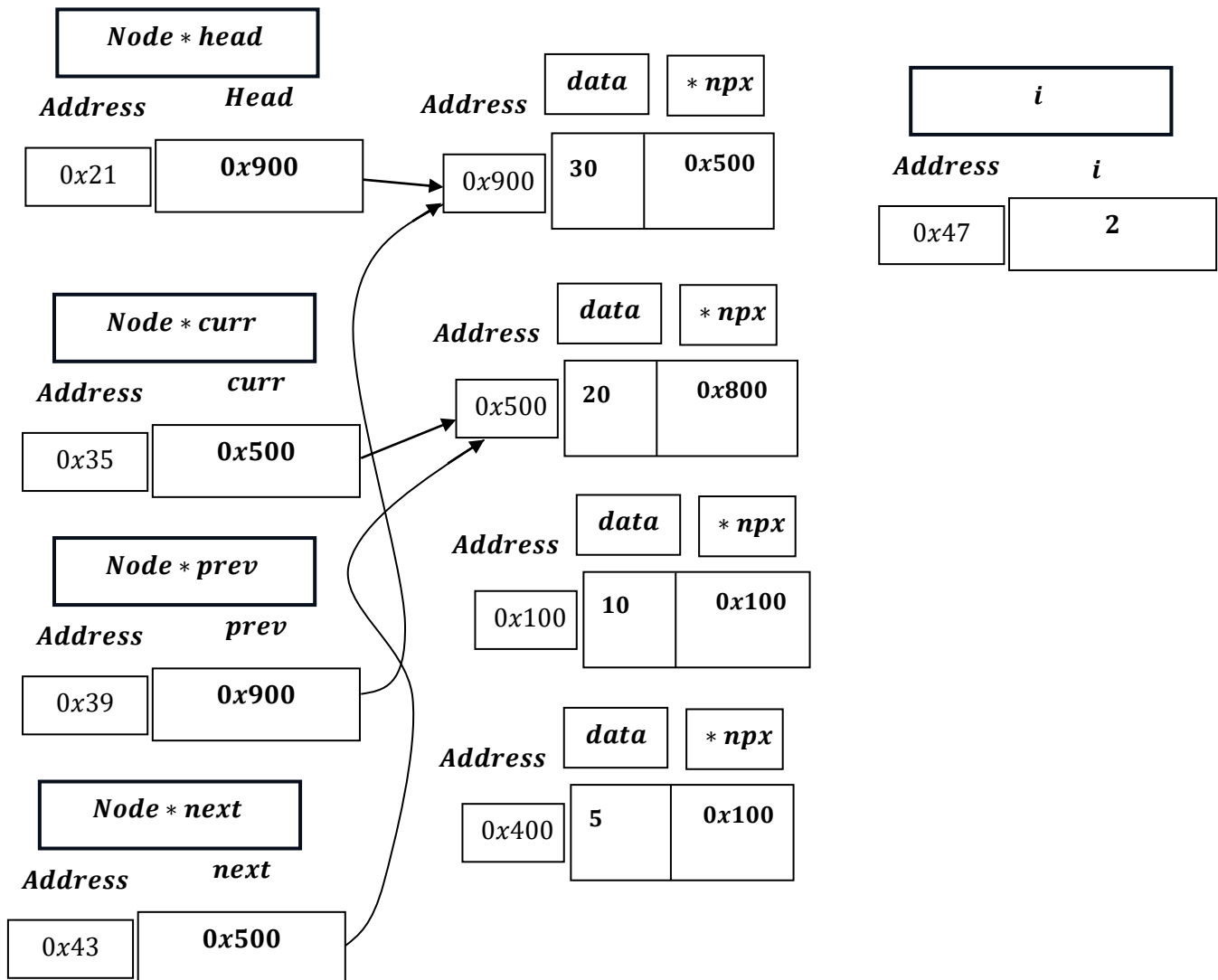
***curr: 0x900* $\neq$ *nullptr: 0x000* AND *i = 1* < *pos: 4* :**

next = XOR(prev: 0x000, curr $\rightarrow$ npx: 0x500) = 0x000 $\oplus$ 0x500 = 0x500.

prev = curr = 0x900

curr = next = 0x500

i = i + 1 = 1 + 1 = 2.



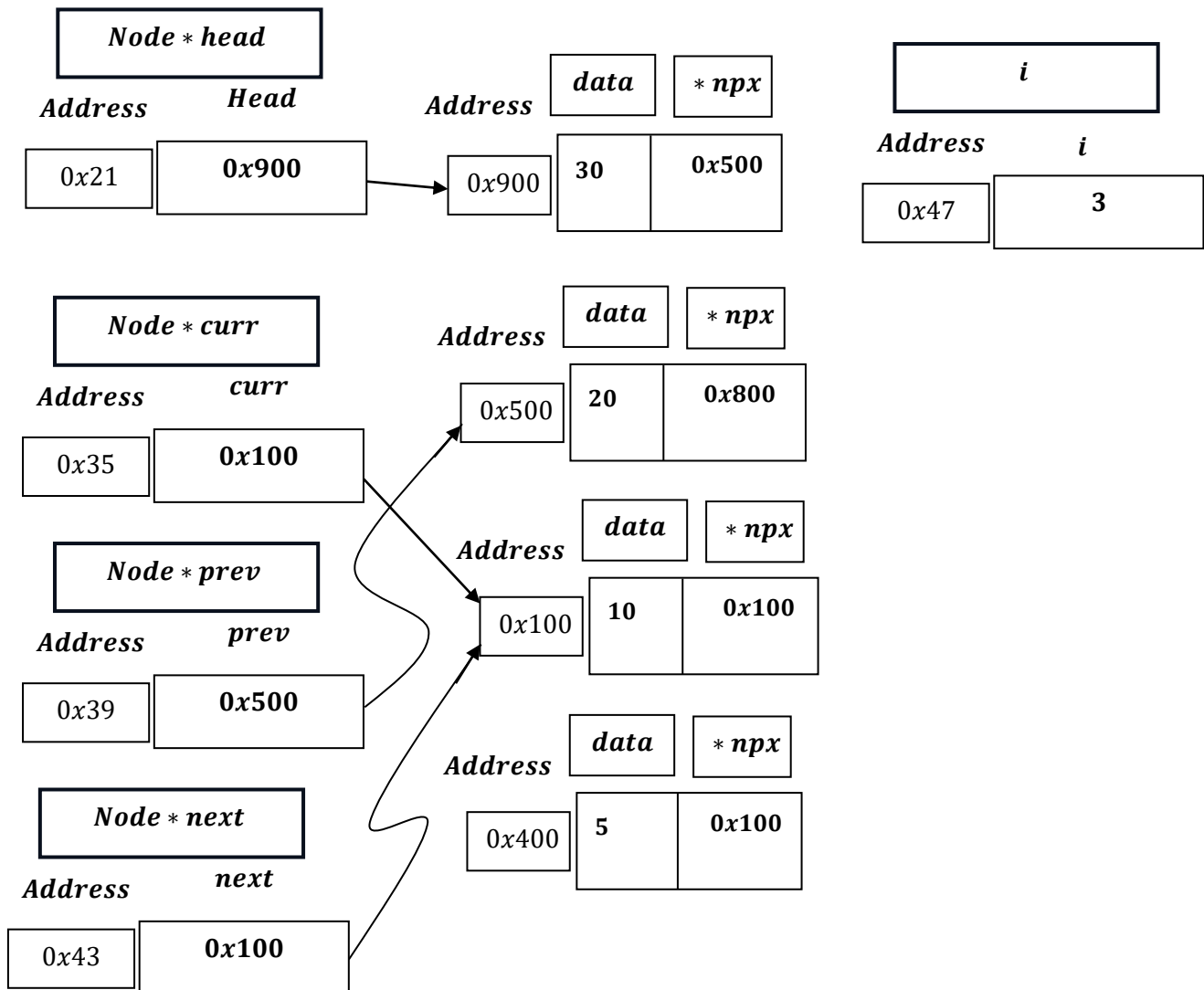
$curr: 0x500 \neq nullptr: 0x000$ AND $i = 2 < pos: 4$:

$next = XOR(prev: 0x900, curr \rightarrow npx: 0x800) = 1001_2 \oplus 1000_2 = 0001_2 = 0x100$

$prev = curr = 0x500$

$curr = next = 0x100$

$i = i + 1 = 2 + 1 = 3.$



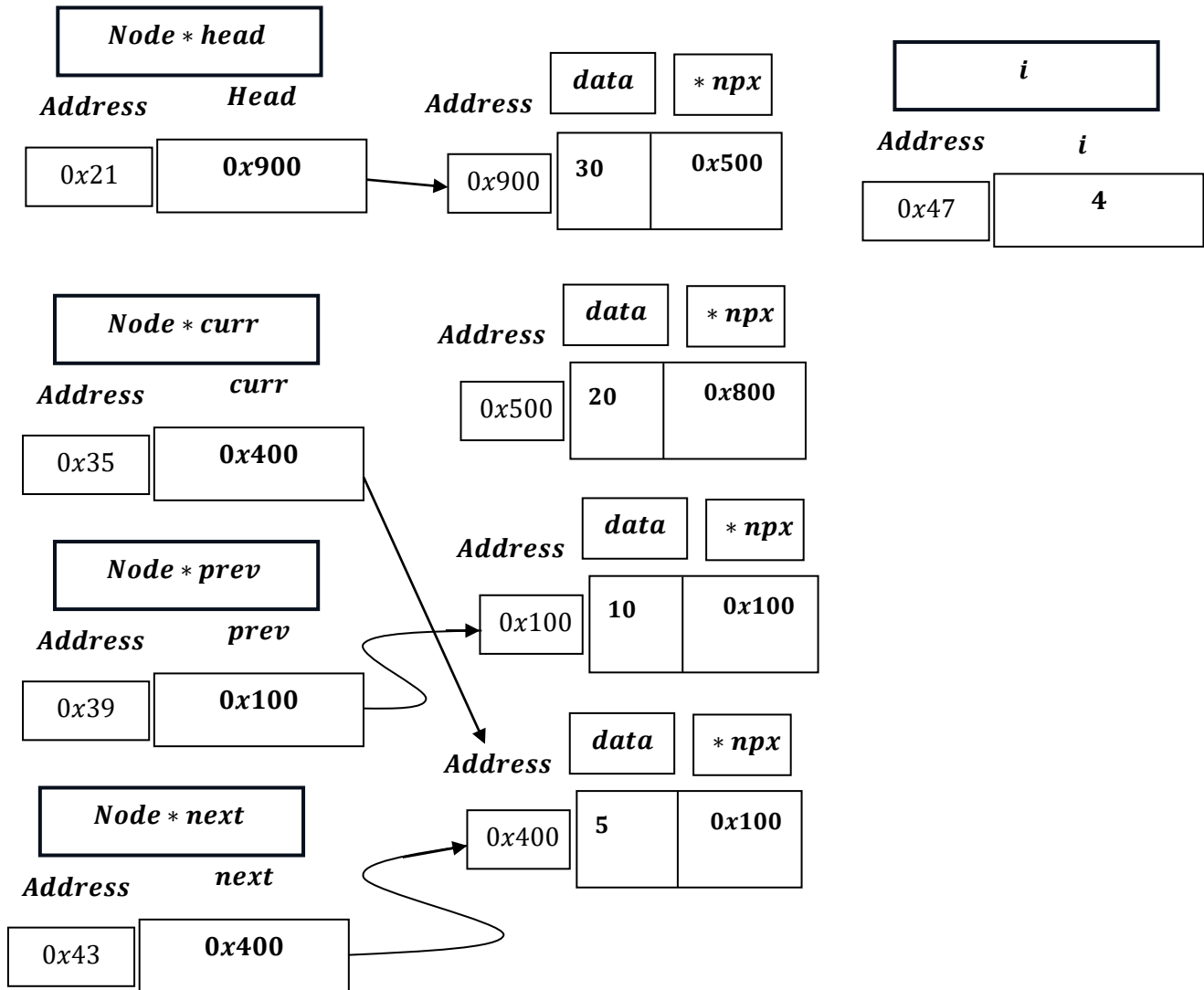
$curr: 0x100 \neq nullptr: 0x000$ AND $i = 3 < pos: 4$:

$next = XOR(prev: 0x500, curr \rightarrow npx: 0x100) = 0101_2 \oplus 0001_2 = 0100_2 = 0x400$

$prev = curr = 0x100$

$curr = next = 0x400$

$i = i + 1 = 3 + 1 = 4.$

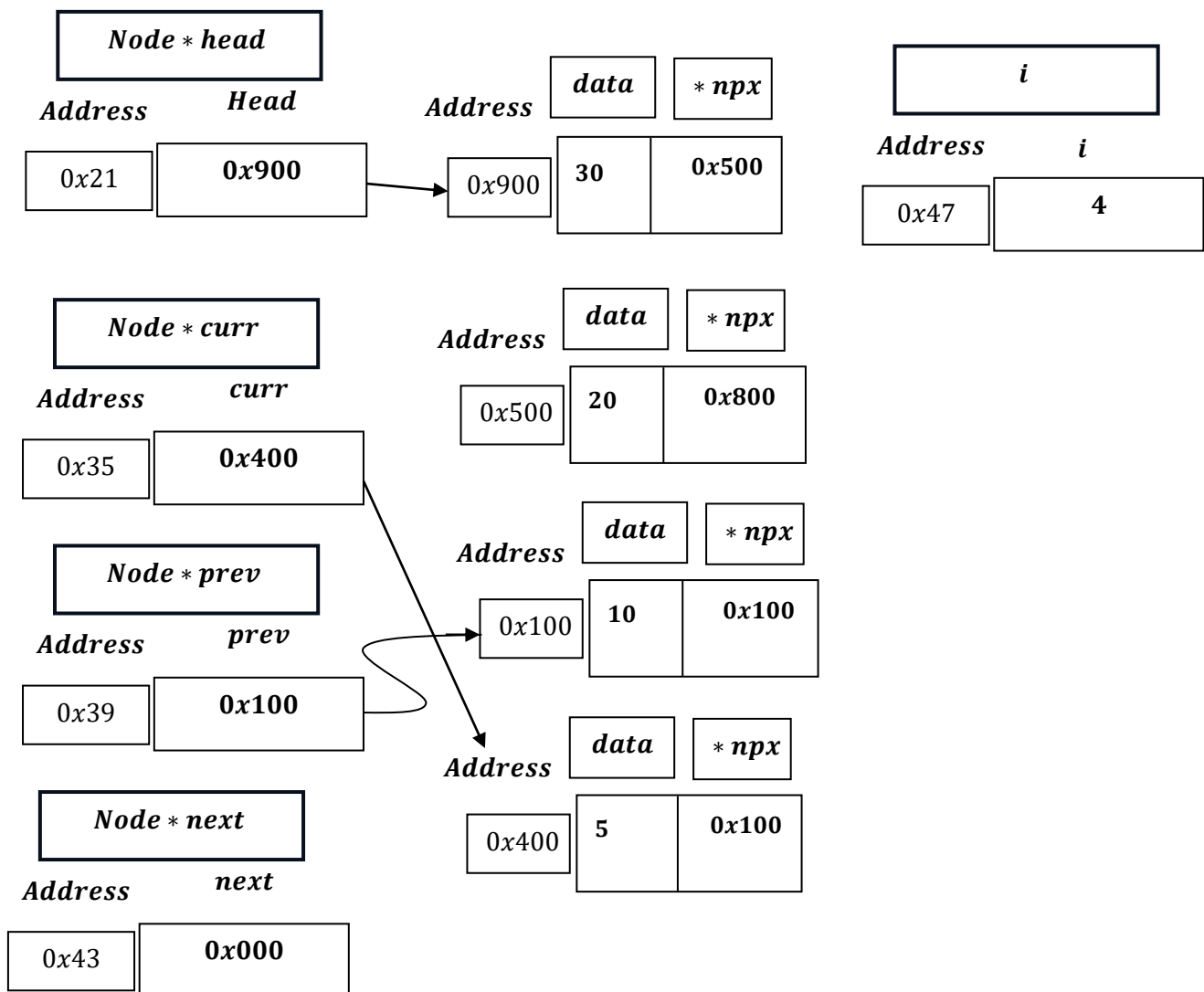


**$curr: 0x400 \neq nullptr: 0x000$ [is TRUE] AND $i = 4 < pos: 4$ [is False]
 $\Rightarrow$ Condition is False.**

```
next = XOR(prev, curr->npx);
```

$next = XOR(prev: 0x100 \oplus curr \rightarrow npx: 0x100);$

$0001_2 \oplus 0001_2 = 0000_2 = 0_{10} = 0x000.$



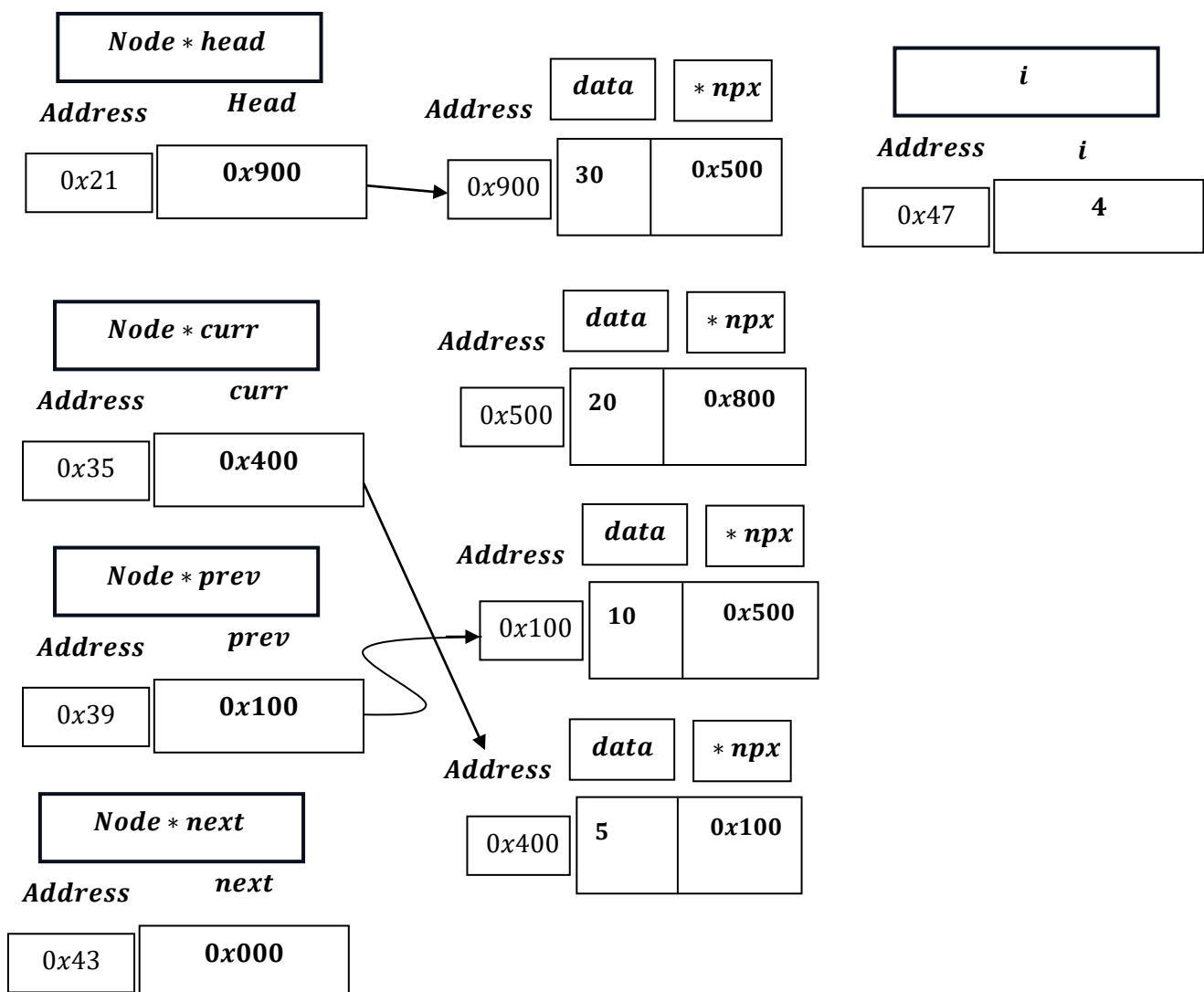
```
Node *prevPrev = XOR(prev->npx, curr);
prev->npx = XOR(prevPrev, next);
```

$prevPrev = XOR(prev \rightarrow npx: 0x100, curr: 0x400)$

$= 0001_2 \oplus 0100_2 = 0101_2 = 5_{10} = 0x500$

$prev \rightarrow npx = XOR(prevPrev: 0x500, next: 0x000)$

$= 0101_2 \oplus 0000_2 = 0101_2 = 5_{10} = 0x500$




```

if (next != nullptr)
{
    Node *nextNext = XOR(curr, next->npx);
    next->npx = XOR(prev, nextNext);
}

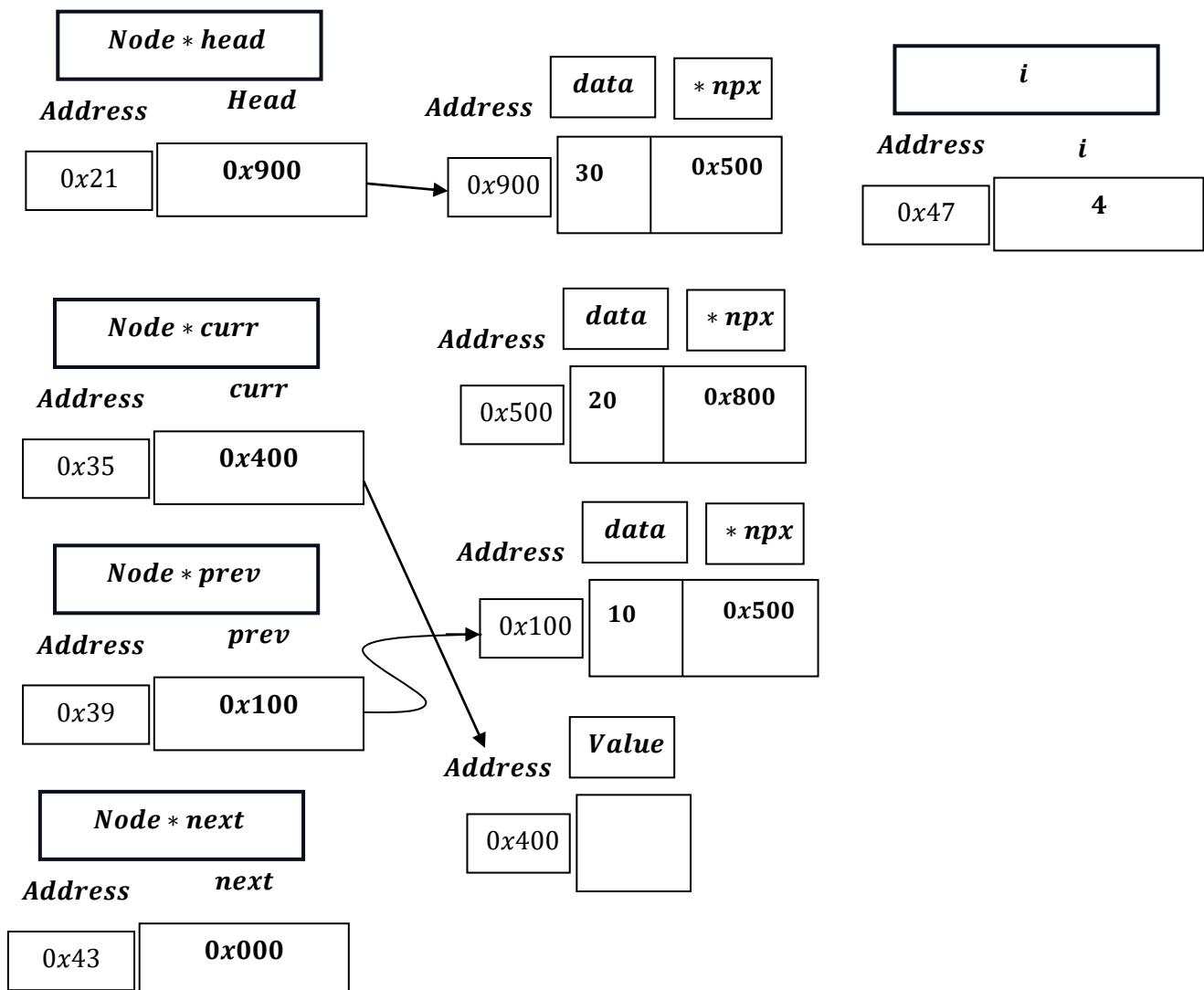
```

if(next:0x000 ≠ nullptr:0x000) → is false.

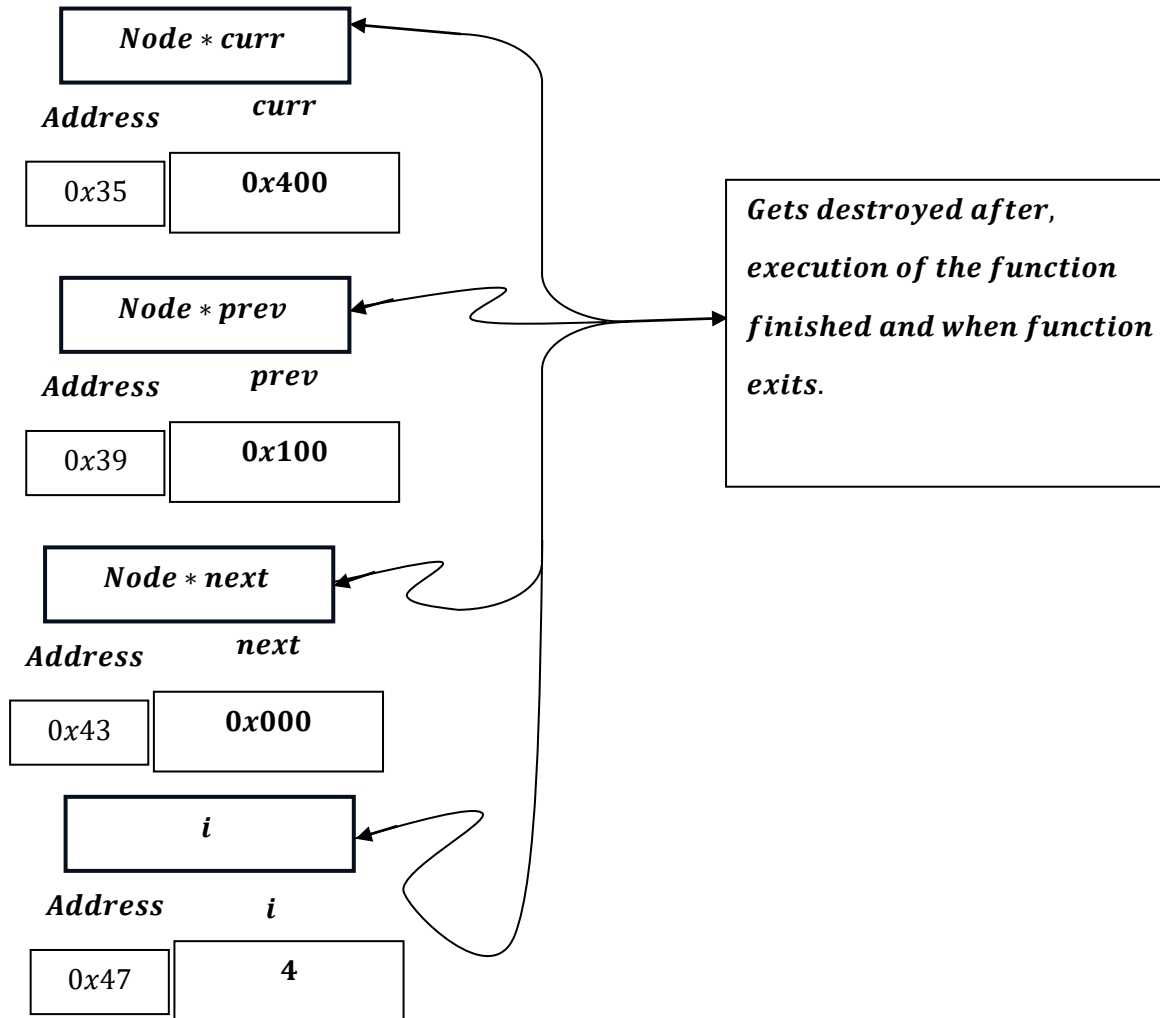
```

free(curr);

```



As , functions ends:



And we are left with:

